

NATURAL LANGUAGE PROCESSING

LESSON 3: POS, SYNTAX, N-GRAMS



OUTLINE

- Part of Speech
 - Types of words and categories
 - Syntax
 - Natural language rules, and context-free grammar
 - N-Grams
 - Probabilistic data for NLP studies
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PART OF SPEECH (POS) TAGGING

Generally it is divided into 3 groups:

- Noun, pronoun, adjective
 - Verb, adverb, auxiliary verb (yardımcı fiil)
 - Preposition (edat), conjunction (bağlaç), and exclamation (ünlem).
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WHY WE NEED POS TAGGING?

For accurate stemming: if we know the class of word we can analyze the suffixes in right scope

- Bu ev mavi renge boyanmış. -> Verb boya(mak) – n (reflexive verb suffix)+(~past perfect continues)
 - Ev için kullanılan senin boyanmış. -> Noun boya + n (2nd person possessive suffix) + (~past perfect continues)
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WHY WE NEED POS TAGGING?

- For quickly finding names or other phrases
 - For finding instances or frequencies of particular constructions in large corpora.
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POS TAGGING CATEGORIES

It has two broad supercategories:

1. Closed class : fixed word types. Rarely new words gained.

- Most of the members are **Function Words**.
 - Ders kayıtlı yaparken **herhangi bir** dersin saati **bu** dersin saati **ile** karşılaşmaması gerekir.

2. Open class: these types can gain new words from derived words or borrowed words from other languages.

Four major types: Noun, Adjective, Verb, Adverb



POS TAGGING CATEGORIES

Nouns:

- Proper Nouns: unique names for people, places, etc. Usually capitalized.
- Common Nouns: general names for everything.

Verbs:

- Verbs: Refer to actions, processes, occurrences. Koşmak, konuşmak, doğmak
 - Auxiliary Verbs: help names to act as a verb group: etmek, -(e)bilmek
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POS TAGGING CATEGORIES

Adjectives: describes the properties or qualities of nouns

- Chinese does not have adjectives
- Turkish has plenty of adjectives with plenty of subclasses

Adverbs:

- Most undetermined class: mostly modifies verbs, adverbs, entire verb phrases.
 - Bu önemli belgeyi, kurum **kapanmadan**, **çok hızlı** **koşarak** yetiştirmelisiniz.
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POS TAGGING CATEGORIES

- Preposition, conjunction and exclamation: members of closed class

Natural languages may have different sets of these closed class part of speech members.

English has more:

prepositions: on, under, over, near, by, at, from, to, with

determiners : a, an, the

particles: up, down, on, off, in, out, at, by

numerals: one, two, three, first, second, third

POS TAGGING CATEGORIES

- Pronouns: They are member of closed class but they act as a kind of shorthand for referring to some noun phrases or entity or event.

- Ali okula geldi mi? Bugün onu göremedim.



PENN TREEBANK TAGS

- PENN Treebank
 - 45 Part-of-Speech tags
 - Operated between 1989-1996
 - approximately 7 million words of part-of-speech tagged text.

```
( (S
  (NP Martin Marietta Corp.)
  was
  (VP given
    (NP a
      $ 29.9
      million Air Force contract
      (PP for
        (NP low-altitude navigation
          and
          targeting equipment))))))
.)
```

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	<i>and, but, or</i>	SYM	Symbol	<i>+, %, &</i>
CD	Cardinal number	<i>one, two, three</i>	TO	"to"	<i>to</i>
DT	Determiner	<i>a, the</i>	UH	Interjection	<i>ah, oops</i>
EX	Existential 'there'	<i>there</i>	VB	Verb, base form	<i>eat</i>
FW	Foreign word	<i>mea culpa</i>	VBD	Verb, past tense	<i>ate</i>
IN	Preposition/sub-conj	<i>of, in, by</i>	VBG	Verb, gerund	<i>eating</i>
JJ	Adjective	<i>yellow</i>	VBN	Verb, past participle	<i>eaten</i>
JJR	Adj., comparative	<i>bigger</i>	VBP	Verb, non-3sg pres	<i>eat</i>
JJS	Adj., superlative	<i>wildest</i>	VBZ	Verb, 3sg pres	<i>eats</i>
LS	List item marker	<i>1, 2, One</i>	WDT	Wh-determiner	<i>which, that</i>
MD	Modal	<i>can, should</i>	WP	Wh-pronoun	<i>what, who</i>
NN	Noun, sing. or mass	<i>llama</i>	WPS	Possessive wh-	<i>whose</i>
NNS	Noun, plural	<i>llamas</i>	WRB	Wh-adverb	<i>how, where</i>
NNP	Proper noun, singular	<i>IBM</i>	\$	Dollar sign	<i>\$</i>
NNPS	Proper noun, plural	<i>Carolinas</i>	#	Pound sign	<i>#</i>
PDT	Predeterminer	<i>all, both</i>	"	Left quote	<i>(' or ")</i>
POS	Possessive ending	<i>'s</i>	"	Right quote	<i>(' or ")</i>
PP	Personal pronoun	<i>I, you, he</i>	(	Left parenthesis	<i>([, { , <)</i>
PPS	Possessive pronoun	<i>your, one's</i>	)	Right parenthesis	<i>([, { , >)</i>
RB	Adverb	<i>quickly, never</i>	,	Comma	<i>,</i>
RBR	Adverb, comparative	<i>faster</i>	.	Sentence-final punc	<i>(. ! ?)</i>
RBS	Adverb, superlative	<i>fastest</i>	:	Mid-sentence punc	<i>(; : ... - -)</i>
RP	Particle	<i>up, off</i>			

Figure 8.6 Penn Treebank Part-of-Speech Tags (Including Punctuation)

SYNTAX

- Language is not a bag of words
- Syntax -> «Setting out together or arrangement»
- Gramatical rules apply to categories and groups of words, not individual words.
- A sentence includes a subject and a predicate. The subject is a noun phrases and the predicate is a verb phrase.
- When people learn a new word, they learn its syntactic usage.
 - See in a sentence, you will easily find out unknown word's category.
 - «Students will be really zealous for this class.»

SYNTAX

- Constituents -> words or word phrases that has solid meaning
 - Each word is a constituent
 - Constituents are non-crossing, any two constituents has an intersection than one of them containing the other.
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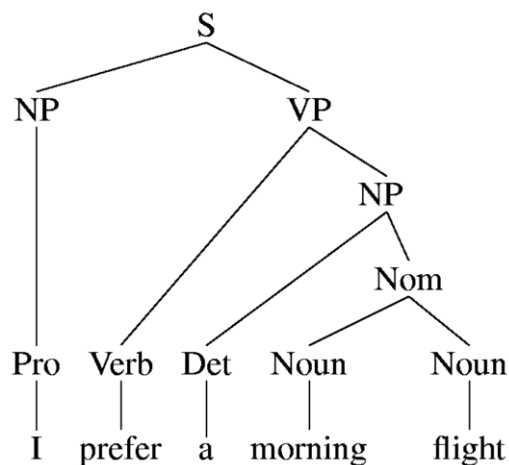
SYNTAX

- Constituents Tests
 - Pronoun test: change constituent candidate with a pronoun if it sounds right then it is correct
 - A small dog is barking in the park -> It is barking in the park. ✓
 - Question test:
 - I have seen blue elephants.
 - What I have seen? -> blue elephants
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CONTEXT FREE GRAMMARS FOR NLP

- English has two main group: Noun Phrase (NP) and Verb Phrase (VP)
 - $S \rightarrow NP VP$
 - $NP \rightarrow DT CN$
 - $NP \rightarrow PN$
 - $VP \rightarrow V \mid NP$
 - $DT \rightarrow a \mid the$
 - $CN \rightarrow child \mid cat \mid dog$
 - $PN \rightarrow Samantha \mid Jorge \mid Min$
 - $V \rightarrow took \mid saw \mid liked \mid scared \mid chased$

CONTEXT FREE GRAMMARS FOR NLP



CONTEXT FREE GRAMMARS FOR NLP

- Wherever N is allowed in a sentence,
 - DT N -> the cat
 - JJ N -> white cat
 - DT JJ N -> the white cat
- are also allowed
- We can use the notation for alternatives
 - NP -> N | DT N | JJ N | DT JJ N
- Optional categories can be also marked using parantheses:
 - NP -> (DT) (JJ) N

CONTEXT FREE GRAMMARS FOR NLP

- Verb Phrases
 - Samanta ran. -> VP -> V
 - Samanta ran to the park. -> VP -> V P NP
 - Samanta ran away. -> VP -> V P
 - Samanta bought a cookie. -> VP -> V NP
 - Samanta bought a cookie for John -> V NP P NP
 - Overall structure: VP -> V (NP) (P) (NP)

CONTEXT FREE GRAMMARS FOR NLP

- Adding Prepositional Phrases (on, under, over, near, by, at, from, to, with) end of Noun Phrases
 - S -> NP VP
 - NP -> (DT) (JJ) N (PP)
 - VP -> V (NP) (PP)
 - PP -> P (NP)
- Whenever a preposition is allowed, it can be followed by a noun phrase.
- NP can contain any number of PPs but only up to two NPs.
- Humans prefer PPs less than 4 in a NP.

CONTEXT FREE GRAMMARS FOR NLP

- The boy saw the woman with the telescope.
 - Did the boy saw
 - The woman that has a telescope with his bare eyes?
 - The woman with his telescope?
- Çocuk kadını teleskopla gördü.
- Acaba çocuk
 - Kadını teleskopla birlikte mi gördü? -> Çocuk, teleskoplu kadını gördü.
 - Kadını teleskoptan mı gördü? -> Çocuk teleskopla kadını gördü.

CONTEXT FREE GRAMMARS FOR NLP

- Word order and flexible inflectional suffix may resolve out most of the ambiguous sentences in English.
- Turkish actually uses Subject – Object – Verb order but in theory it has a free word order for sentence. This makes the language use constituent orders to emphasize the info in the sentence.
- Any phrase or word before predicate has the emphasized information for that sentence.

WORD ORDER

Word order	English equivalent	Proportion of languages	Example languages
SOV	"She him loves."	45% 	Proto-Indo-European, Sanskrit, Hindi, Ancient Greek, Latin, Japanese, Korean
SVO	"She loves him."	42% 	English, French, Hausa, Indonesian, Malay, Mandarin, Russian
VSO	"Loves she him."	9% 	Biblical Hebrew, Arabic, Irish, Filipino, Tuareg-Berber, Welsh
VOS	"Loves him she."	3% 	Malagasy, Baure, Proto-Austronesian
OVS	"Him loves she."	1% 	Apalaí, Hixkaryana
OSV	"Him she loves."	0%	Warao

Frequency distribution of word order in languages surveyed by Russell S. Tomlin in 1980s^{[10][11]} (V · T · E)

WORD ORDER

■ Adjective Ordering in English

- Det
- Number
- Strength
- Size
- Age
- Shape
- Color
- Origin
- Material
- Purpose
- Noun

■ Adjective Ordering in Turkish

- Number
- Personal Opinion
- Size
- Age
- Shape
- Color
- Origin
- Material
- Purpose
- Noun

There is no strict order. It is unusual to use 4 or more adjectives for a noun.

N-GRAMS

■ Why we need them?

- They are leaving in about fifteen *minuets* to go to her house.
- He is trying to *fine* out.

If we know the probabilities of

minute and minuet with other words, or find and fine

We can assume **fifteen minute** or **find out** is more sensible in the context of the sentences.

N-GRAMS

- Probabilities are based on counting things and Natural Language Processing works with **words** as countable data.
 - Bigram -> looks one word into the past
-> first-order Markov Model
 - Trigram -> looks two words into the past
-> second-order Markov Model
 - N-gram -> looks N-1 words into the past
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N-GRAMS MODELS

- Can be trained by counting and normalizing
- For bigrams, a particular bigram can be calculated as

$$P(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)}{\sum_w C(w_{n-1}w)}$$

- Can be simplified as:

$$P(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)}{C(w_{n-1})}$$


N-GRAMS MODELS

- Bigram counts for 7 of the words (out of 1.616 total word types) in Berkeley Restaurant Project Corpus of ~10.000 sentences.

	I	want	to	eat	Chinese	food	lunch
I	8	1087	0	13	0	0	0
want	3	0	786	0	6	8	6
to	3	0	10	860	3	0	12
eat	0	0	2	0	19	2	52
Chinese	2	0	0	0	0	120	1
food	19	0	17	0	0	0	0
lunch	4	0	0	0	0	1	0

N-GRAMS MODELS

- from Turkish Dictionary

2-Gram	olma durumu	4359
	-Hayırsız olma durumu	
	-Uçarı olma durumu	
	bir biçimde	1196
	-Çekimsere yakışır bir biçimde	
	-Tedbirsiz bir biçimde, tedbirsiz olarak	
	yaptığı iş	907
	-Telgrafçının yaptığı iş	
	-Kapıcının yaptığı iş	

N-GRAMS MODELS

■ from Turkish Dictionary

3-Gram	işine konu olmak	416
	-Başlama işine konu olmak	
	-Aktarma işine konu olmak	
	Bu renkte olan	214
	-Bu renkte olan	

N-GRAMS MODELS

■ from Turkish Dictionary

4-Gram	ihtimali veya imkânı bulunmak	1788
	-Yavaşlama ihtimali veya imkânı bulunmak	
	-Tutulma ihtimali veya imkânı bulunmak	
	iline bağlı ilçelerden biri	1061
	-Adana iline bağlı ilçelerden biri	
	-Ankara iline bağlı ilçelerden biri	
	yapan veya satan kimse	202
	-Tatlı yapan veya satan kimse	
	-Yoğurt yapan veya satan kimse	

JUST GUESS: QUESTION

How about last one from me?

- If we look appearance in English written books from 1800 to 2000 for 3 N-Grams: ['Alber Einstein', 'Sherlock Holmes', 'Frankenstein'] what will be the graph of these three N-Grams?
- Hint:
 - Sherlock Holmes first appears in *A Study in Scarlet*, 1887
 - Frankenstein first published at 1818
 - Alber Einstein published his paper about general relativity at 1916 and win Nobel Prize of Physics at 1921
- Cheat: try <https://books.google.com/ngrams>

JUST GUESS: ANSWER

